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School of Computing, Information and Mathematical Sciences

CS211: Computer Organization
Mid-Semester Test – Semester 2, 2012

Time Allowed: 1 hour 40 mins
Total Marks: 55 (20% of final grade)

Name: _____ Student ID: _____

INSTRUCTIONS:

- There are 10 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

a) What is the difference between half-adder and full-adder? (2 marks)

Half-adder does not include any carry-in bit (1 mark)
Full-adder includes carry-in bit (it is made of two half adders) (1 mark)

b) What is one major problem that exists in the Clocked SR-Latch? (1 mark)

An error (ambiguity) occurs in the SR latch when S and R inputs are both 1

c) How does a Clocked D-latch solve the above problem? (1 mark)

It prevents both inputs S and R being 1 at the same time by providing one input D and its complement

d) List three different ways to achieve processor –level parallelism when designing computers. (1 mark)

- Array computers
- Multi processors
- Multi computers

Question 2

a) What is the largest positive number that can be represented in 10 bits using one's complement? Represent the number in bits and decimal as well. Assume the first bit is the sign bit. (2 marks)

0111111111 = 511

b) What is the smallest negative number that can be represented in 10 bits using one's complement? Represent the number in bits and decimal as well. Assume the first bit is the sign bit. (2 marks)

1000000000 = -511

c) Represent -18 in 8 bit binary using two's complement. (2 marks)

18 = 00010010
-18 = 11101101 + 1 = 11101110

Question 3

a) Show the hexadecimal storage for the largest possible IEEE single precision floating point number. Justify your answer. (3 marks)

Largest number has to be positive (thus sign bit of 0), exponent cannot be all 1s (this represents infinity, thus largest valid value would be 11111110 i.e. one less than all 1s) and largest fraction component can be all 1s, thus:
0 11111110 11...11
In hex this is: 7F7FFFFF

b) Convert -19.3 to IEEE single-precision format. Represent the final answer in hexadecimal. (4 marks)

19 = 10011
0.3 = 0.010011……..
19.3 = 10011.010011…….
Normalize = 1.0011010011… X 2^4

Stored exp = actual + excess
= 4 + 127
=10000011

IEEE : 1 10000011 001101001100110 = C19A6667

Question 4
For the truth table given below:

x	y	z	F(x,y,z)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

a) Write the disjunctive normal form for the above truth table. (2 mark)

ANS: $x\bar{y}\bar{z} + x\bar{y}z + xy\bar{z} + xyz + \bar{x}yz$

b) Evaluate the boolean expression in part a) above in its simplest form and draw the circuit diagram. (5 marks)

Final answer: x + yz (1 mark for working; 2 marks for final answer; 2 marks for circuit diagram)

Question 5

What would be the code word if you encode the data word 10100011 to allow one bit error detection and correction using hamming code with even parity? (4 marks)

ANS: 011001000011 (4 marks for correct solution; partial marks for partially correct solution)

Question 6

A magnetic disk has 1000 cylinders, 10 tracks per cylinder, 128 sectors per track, 1 KB per sector. The rotation time is 6 msec, maximum seek time is 150 msec, switching between adjacent cylinders takes 15 msec, and switching between adjacent tracks within a cylinder takes no time. Assume this disk has 4 MB of data stored on it.If data is stored on consecutive tracks and is twice as likely to start on the first half (upper 5 tracks) of a cylinder than the second half (lower 5 tracks), find the average access time. (5 marks)

#Tracks: 4 MB x 1024 KB/1 MB x 1 sector/1 KB x 1 track/128 sectors = 32 tracks
Transfer time = 32 x 6 = 192 msec

Exp. switch and wait:

Probability (twice as likely) → If x = probability, 5 x 2x + 5 x 1x = 1 → x = 1/15

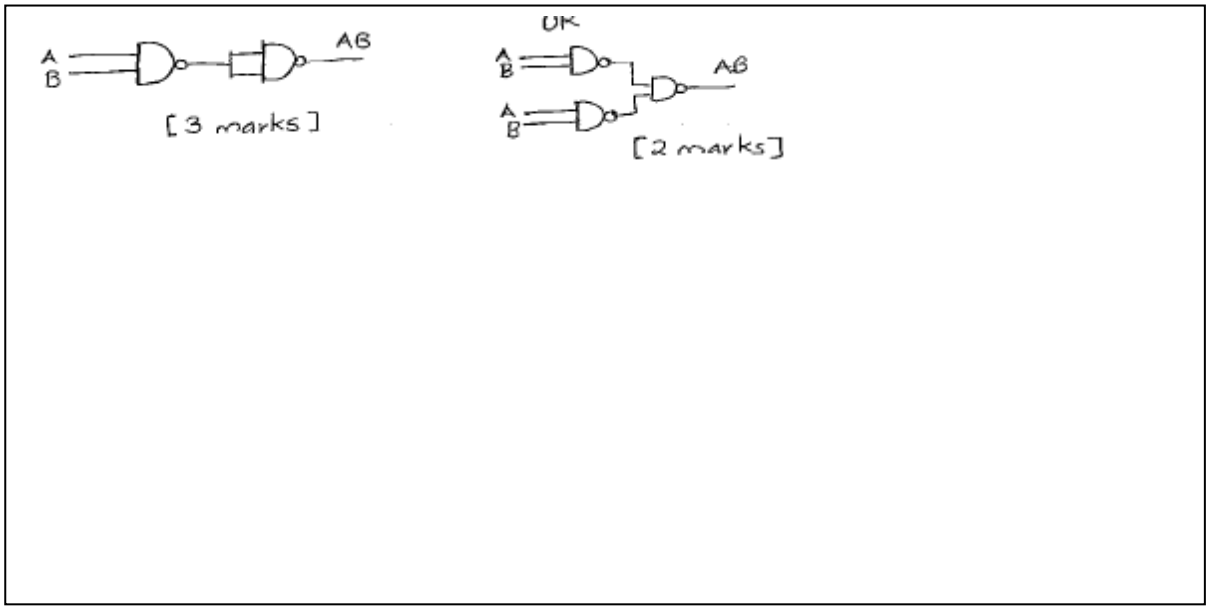
Track	Probability	# switch	s/w time
1	2/15	3	18x3
2	2/15	3	18x3
3	2/15	3	18x3
4	2/15	3	18x3
5	2/15	3	18x3
6	1/15	3	18x3
7	1/15	3	18x3
8	1/15	3	18x3
9	1/15	3	18x3
10	1/15	4	18x4

Exp. s/w = 5 x [(18x3) x 2/15] + 4 x [(18x3) x 1/15] + [(18x4) x 1/15] = 55.2 msec

Exp. access time = exp seek + exp latency + transfer + exp s/w
= (1/3 x 150) + (1/2 x 6) + 192 + 55.2
= 300.2 msec

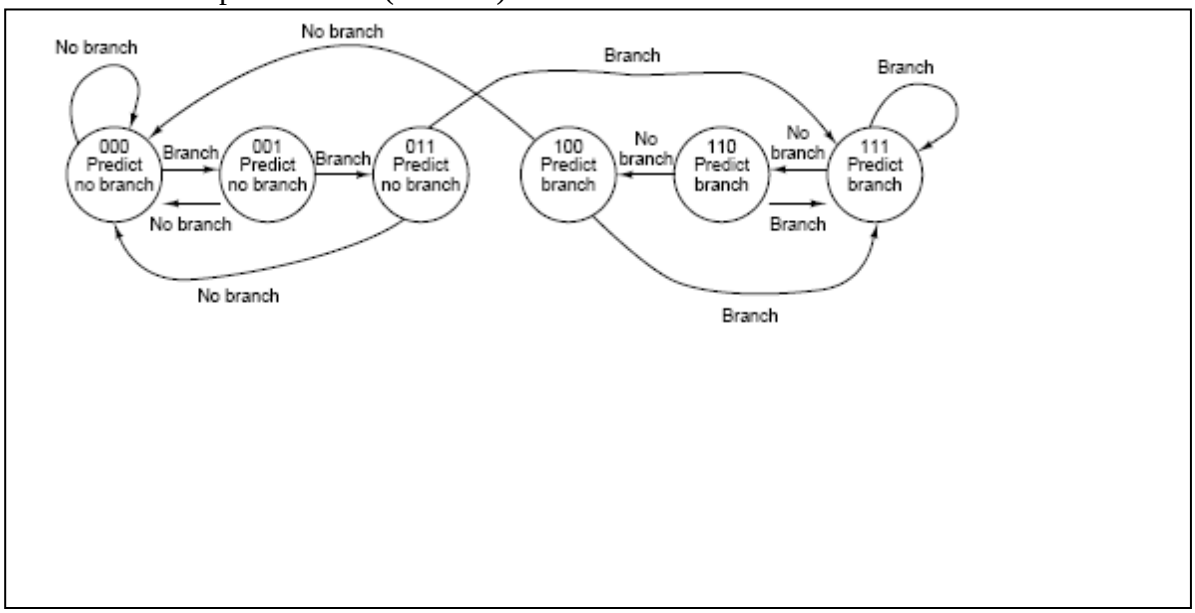
Question 7

Draw a circuit diagram for the AND function on two 1-bit inputs (A and B) using only NAND gates (use minimum number of gates) (3 marks)



Question 8

Draw a finite state machine for branch prediction that changes prediction after three consecutive mis-predictions? (3 marks)



Question 9

Write the output for the following machine language program if the inputs are 6, 3, 4 and -1. The inputs are in sequential order. Show the accumulator value for each input and any changes to it done by the program. (5 marks)

```
0011000000000000
1011000000001010
0101000000010000
0010000000010000
1001000000000000
0001000000010000
0100000000000000
0000000000000000
0000000000001010
```

ANS: 23

Question 10 (10 marks)

Write machine language program which takes two integers as input and should compute the division of integers (first integer read in divided by second integer read). Your program should check that second integer read should not be 0 as division by 0 is not allowed. If the second integer read is 0, ask user to enter another value for second integer. This process should continue unless and until the user enters an integer apart from 0. The program should output the division of two numbers. Do not write the program using binary numbers use decimal numbers instead. For example, for the machine instruction 0011000000000000, write "3 0" instead of 0011000000000000.

```
3     0     ; read first integer
2     18    ; store at loc 18
3     0     ; read second integer
10    4      check if acc = 0, jump to loc 4
2     20    ; store at loc 20
1     18    ; load first integer to acc
8     20    ; divide first integer by second
4     0     ; write
0     0     ; stop
0     0     ; first storage
0     0     ; second storage
```

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